UNIT IV

ASSIGNMENT

Class #6: Properties of eigenvalues and Eigenvectors

1) If $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 5 \\ 0 & 0 & 3 \end{bmatrix}$, then find the eigen values of A^{-1}.

2) If $A = \begin{bmatrix} -1 & 2 & 3 \\ 0 & 3 & 5 \\ 0 & 0 & -2 \end{bmatrix}$, then Find the eigen values of A^2.

3) Find the eigenvalues and eigenvectors of A, A+10I.

$$A = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$$

Ans: Eigen values are 1,5

Eigen Vectors are

$$\begin{pmatrix} -1 \\ 1 \end{pmatrix} k_1$$

$$\begin{pmatrix} 3 \\ 1 \end{pmatrix} k_2$$

Eigen value of $A + 4I$: 11,15